

Activity 2 · Part I

Digital Economy Intelligence

From public data to a comparative diagnosis of the digital economy

Purpose

Build a comparative diagnosis of the digital development of five economies using official data, integrating the acquisition, preparation, analysis, visualization and interpretation of indicators. The analysis must make it possible to identify Mexico's strengths, gaps and limitations relative to the other assigned economies.

Team organization

Team	Members	Assigned countries	Sector
Team 1	6	Mexico, South Korea, Finland, Brazil and India	Telecommunications
Team 2	5	Mexico, United Kingdom, Estonia, Chile and Colombia	Health
Team 3	5	Mexico, Netherlands, Kenya, Argentina and New Zealand	Agriculture
Team 4	5	Mexico, United States, Germany, China and Japan	Science
Team 5	5	Mexico, Norway, United Arab Emirates, Australia and Spain	Energy

Mexico is mandatory for all teams. Each group will make the comparison against a different reference set.

Official sources and access links

All data used must come from official and verifiable sources. It is recommended to start with the following platforms:

Source	Suggested use	Access
ITU DataHub	Official indicators on connectivity, infrastructure and ICT use.	Visit source
ITU - ICT Development Index 2026	Reference page for the index and methodological materials.	Visit source
World Bank Indicators API	Programmatic access to the World Development Indicators and other World Bank databases.	Visit source
World Bank Open Data	Portal for browsing and downloading indicators; lets you review series and download CSV, XML or Excel.	Visit source
UNCTADstat Data Centre	Digital economy, ICT services, digitally deliverable services, e-commerce and technology.	Visit source
UNCTAD - Digital Economy and Technology	Thematic collection of digital economy indicators and insights.	Visit source
WIPO IP Statistics	Official statistics on intellectual property, patents and technology fields.	Visit source
WIPO IP Statistics Data Center	Browse and download historical and recent intellectual property data.	Visit source

Provenance rule

Every value used must be traceable to its organization, indicator, year and unit. If a data point is not available, it must not be replaced by an invented or estimated value without an explicit and justifiable methodology.

Rules for working with data

- Synthetic, invented or manually filled-in data are not allowed without a documented methodology.
- Acquisition must be done via API, official CSV/XLSX download, or programmatic reading of files published by the organization.
- Original files must be kept unmodified in the data/raw/ folder.
- Every transformation applied to the data must be documented in the notebook.
- The observation year must be kept for each indicator. Years must not be mixed without reporting it.
- The analysis must clearly distinguish between a value equal to zero and a data point that is not available.

Challenge

How prepared is each economy to participate in, generate and capture value within the digital economy, and what are Mexico's main strengths and gaps relative to the other four economies on the team?

1. Data acquisition and integration

- Obtain the necessary information from the authorized official sources.
- Document the download or query procedure so that another person can reproduce it.
- Keep original files and processed files separate.
- Generate an integrated dataset containing exclusively the five assigned economies and the eight selected indicators.

2. Mandatory indicator selection

- Each team must use exactly eight indicators. Analyses with fewer or more indicators will not be accepted.
- The distribution will be: 2 infrastructure or connectivity indicators; 1 access or use; 1 quality or affordability; 2 digital economic activity; 2 technological capacity or innovation.
- All eight indicators must exist for the five economies or, failing that, the team must document the absence and justify the treatment of the missing data.
- Taken together, the eight indicators must come from at least three of the official sources listed in this document.

3. Recording and traceability

- For each indicator, record: official name, definition, unit, source, source link, observation year, acquisition method and transformation applied.
- The record must allow any figure in the analysis to be traced back to its original source.

4. Temporal comparability

- For each indicator, use the most recent year that allows a reasonable comparison across the five economies.
- Explain what problems can arise when comparing indicators from different years and when the comparison could be misleading.

5. Exploration and preparation

- Identify missing values, scales, variable types, extreme values and inconsistencies.
- Document all decisions made during preparation.
- Do not hide missing data or automatically replace values without justifying the procedure.

6. Conceptual foundations

A. ICT sector, digital economy and digitalized economy

Explain the difference between the three concepts. Then indicate which one best describes the set of indicators used in your analysis and justify your answer.

B. Growth of the digital economy

Select three of your eight indicators and explain which different dimension of digital growth each one represents. Explain why a single indicator is not enough to claim that an economy is digitally developed.

C. Indicator, evidence and interpretation

Choose one indicator and develop the sequence: observed data → interpretation → supported conclusion. Then state a conclusion that could not be justified by that data point alone.

D. Digital trade

Explain the difference between an e-commerce transaction and a digitally delivered service. Include an example of each.

7. Digital Readiness Score (DRS)

Build a composite indicator using the eight indicators. The team must decide and justify the normalization method, the direction of each variable, the weights and the treatment of missing data.

$$DRS = w_1I_1 + w_2I_2 + \dots + w_8I_8$$

The weights must add up to 1. If an indicator has an inverse direction (for example, a higher relative cost implies lower affordability), you must adjust its treatment before integrating it into the index.

8. Data visualization

- Build at least three different visualizations.
- Each visualization must answer a different analytical question.
- At a minimum, they must show: the relative position of the five countries; Mexico's strengths and gaps; and differences between digital profiles that are not visible through the DRS alone.
- Below each visualization include: What do I observe? What does it mean? What can I not conclude?

9. Complementary analysis

Option A. Relationships between variables

Explore relationships between indicators using correlations, scatter plots or another justified technique. Identify at least one relevant relationship and explain what additional evidence would be needed to defend a causal relationship.

Option B. Clustering

Apply a clustering technique to identify digital profiles. The result must be interpreted: describe which characteristics distinguish each group and what the clustering adds compared with the DRS ranking.

10. Digital divide

Based on your results, identify Mexico's main digital gap relative to the other four economies. Analyze the problem considering availability, access, affordability, quality, capabilities and outcomes. Explain why high network coverage does not necessarily imply a high level of use or uptake.

11. Readiness for Artificial Intelligence: the 4C framework

- **Connectivity:** infrastructure, networks and access.
- **Compute:** computing capacity, cloud, data centers and technological infrastructure.
- **Context:** data, languages, applications and resources suited to the context.
- **Competency:** skills to adopt, adapt and develop technology.

Map the available indicators to these dimensions and answer: is the information obtained sufficient to claim that any of the countries is prepared to develop and use advanced AI systems? If not, indicate what additional information would be needed.

12. Final diagnosis

In a maximum of 250 words, answer:

1. How does Mexico compare with the other economies?
2. What is its main strength?
3. What is its main gap?
4. Which country is the most interesting point of comparison, and why?
5. What is the main limitation of the dataset used?

Part I deliverables

- Reproducible notebook: Digital_Economy_Analysis.ipynb.
- Processed dataset: digital_economy_clean.csv.
- Source Log with the provenance of the eight indicators.
- Data Dictionary with the definition of the variables.
- Single-page dashboard or main screen.
- Final diagnosis integrated into both the notebook and the dashboard.

Minimum working structure

TeamXX_DigitalEconomy/

```
data/raw/
data/processed/
notebooks/Digital_Economy_Analysis.ipynb
dashboard/
source_log.csv
data_dictionary.csv
```

Team responsibility

Although the work is done as a team, any member must be able to explain the provenance of the data, the meaning of the indicators, the treatment applied, the DRS calculation and the conclusions reached. Technical and conceptual decisions are part of the evaluation. Therefore, each team member must submit the work individually; otherwise, they will automatically receive a grade of ZERO.

Evaluation criteria

Criterion	Points
Acquisition, traceability and source documentation	1.5
Data preparation and quality	1.0
Selection and understanding of the eight indicators	1.5
Conceptual foundations	1.5
Digital Readiness Score	1.5
Analysis and visualization	1.5
Diagnosis, gap and limitations	1.5
Total	10 points